ANDREEA POCOL

Ph.D. candidate, Computer Science University of Waterloo Website apocol[at]uwaterloo[dot]ca apocol AndreeaPocol Google Scholar

EXPERIENCE

Instructor
WatSPEED

Feb 2026 – Apr 2026 Waterloo, ON

- Course: *Python 1*
- Topics: Functions, Variables and Conditional Statements, Input/Output, Mutation, Strings, Tuples, Iteration, Lists, Sets, Dictionaries and Comprehension, File Input and Output, Classes
- Delivered live and digital course content while facilitating technical orientation and Q&A sessions
- Moderated discussion forums and addressed student inquiries within a 24-hour window to ensure high satisfaction

Course Supporter
WatSPEED

Nov 2024 – Feb 2025 Waterloo, ON

- Course: *Secure Coding*
- Languages: Python, C
- Topics: Architecture, Design & Implementation; Secure Coding with AI; Fuzzing, Profilers, & Penetration Testing
- Responsibilities: Graded assessments, provided individualized feedback to learners

Teaching Assistant, Instructional Assistant
University of Waterloo

Sep 2022 – Dec 2022 Waterloo, ON

- Held office hours, answered questions, marked assignments
- CS 116 (Intro to Computer Science) - W23, F23, F24, W25, S25, F25, W26
- CS 492 (Social & Ethical Implications of Computing) - S23, W24, S24
- CS 488/688 (Computer Graphics) - F22
- CS 350 (Operating Systems) - F21

Software Developer
Magnet Forensics

Dec 2021 – Aug 2022 Waterloo, ON

- Improved cloud build stability, upgradeability & scalability
- Used **CloudFormation** to provision our infrastructure as code, **S3** to store configuration files / templates, **EC2** to spin up applications, **ECS** & **ECR** to manage **Docker** containers, **CloudWatch** to troubleshoot applications, **Route 53** to create health checks and alarms, & **Lambda** to monitor and maintain build fleets (tested using **serverless**)
- Set up a **Grafana** dashboard to monitor security vulnerabilities and issues, pulling in metrics using **Whitesource** and AWS **SecHub** (which feeds data into **EventBridge** and then **CloudWatch**), and presented the project at a company hackathon

Software Developer
Ubisoft

Sep 2020 – Apr 2021 Toronto, ON

- Conducted R&D for a novel internal performance capture tool
- Developed, tested and peer-reviewed code in a **C#.NET / XAML (WPF)** frontend and **Python** backend

Software Developer
Oracle Corporation

Jan 2020 – Apr 2020 Kitchener, ON

- Designed and built features for both new and legacy software in **Java**, and conducted unit testing using the **JUnit** framework

SKILLS

Python | C++ | SQL | Bash | Object-oriented programming | OpenCV | Unit testing | Command line (Linux) | Algorithm design & efficiency | Algorithm analysis | Algorithmic design paradigms | Stereo 3D

ACTIVITIES

Research Assistantship
Dr. Lesley Istead

May 2021 – Aug 2021 Waterloo, ON

- Read papers, wrote **Python** code, looked for connections between colour palettes and film genre, director, etc., wrote paper sections

Research Assistantship
Dr. Lesley Istead

May 2020 – Aug 2020 Waterloo, ON

- Read papers, wrote **OpenCV**, **C++** code, worked on improving edge extraction methodology for stereoscopic 3D line drawing, produced result images, wrote paper sections, assisted in user study

Research Assistantship
Dr. Lesley Istead

Sep 2019 – Dec 2019 Waterloo, ON

- Read papers, wrote **OpenCV**, **C++** code, generated results

Academic Service – External Reviewer
ACM ISS (PACM HCI) 2025 & 2026

Sep 2025, Mar 2026 Remote

- Performed peer review for 2 papers, evaluating submissions on technical novelty, empirical validity, and contributions to the field

Open Source Contribution
jenkinsci/nomad-plugin

July 2019

github.com/jenkinsci/nomad-plugin

- Enhanced the **Nomad** plugin for **Jenkins** (**Java**) with new features, driver support, and improved **Docker** integration

HONOURS & AWARDS

- **Natural Sciences and Engineering Research Council (NSERC) PGS D**, University of Waterloo (2023-2026)
- **The Joseph Wai-Hung Liu Graduate Scholarship**, University of Waterloo (2023, 2024)
- **OGS/QEII-GSST** (declined), University of Waterloo (2023)
- **President's Graduate Scholarship**, University of Waterloo (2023, 2024)
- Graduate Research Studentship, University of Waterloo (2021, 2022, 2023)
- Math Domestic Graduate Student Award, University of Waterloo (2021, 2022, 2023, 2024)
- University of Waterloo Graduate Scholarship (2021, 2022, 2023)
- President's Research Awards, University of Waterloo (2021, 2020)
- University of Waterloo-Waterloo County Entrance Scholarships (2017, 2018)
- President's Scholarship of Distinction, University of Waterloo (2017)
- **Governor General's Academic Medal**, Former Governor General of Canada David Johnston (2016)
- **Dr. Mabel B. Dunham Award**, Grand Valley Institute for Women (2016)

- Developed frontend logic in **JavaScript** and UI / UX with **CSS**
- Developed code in a virtual development environment built on **Oracle SQL Developer**, **Vagrant** and **Oracle VirtualBox**
- Developed robust, browser-based, **GUI tests** using **Groovy**, **XPATH**, **HTML**, **Java** and **Selenium Webdriver**

Software Tools & Infrastructure Developer

Christie Digital Systems

May 2019 – Aug 2019 | Kitchener, ON

- Re-architected the engineering infrastructure into a distributed, parallel compute cluster in a **R&D** / **POC** project
- Used **Docker** and **LXD** together with configuration management / provisioning tools - **Packer** (**Groovy**), and **Ansible** (**YML**), monitoring tools (**InfluxData** stack), and binary repository tool **Artifactory** to fully containerize FPGA and software builds for projectors in adherence to the principles of **infrastructure-as-code**

Application Developer

Sun Life Financial

Sep 2018 – Dec 2018 | Waterloo, ON

- Developed **PowerShell** scripts to automate server deployment & installation prep (start & stop services, sites, IIS, COM+, validation)
- Used **Application Life Cycle Management** and **TFS** in projects to develop, code, **unit test** and debug **SQL** Stored Procedures
- Developed and presented a scalable **C#.NET** app which communicates with MS SQL Server and the Event Viewer to monitor programs for crashes and automate issue resolution / escalation which is traditionally handled by on-call personnel
- Created **C#.NET Core** DLLs (e.g. use JSON to post to JIRA through web **API**) for micro-services in a new distributed logging framework

Distributed Developer

Sun Life Financial

Jan 2018 – Apr 2018 | Waterloo, ON

- Developed, coded, tested, and debugged **Visual Basic** scripts towards the maintenance of web-based applications
- Used **SQL**, **T-SQL** and VBA to maintain company **databases** in production via problem analysis, resolution, and reporting
- Interfaced with system designers, business analysts, and application analysts using **JIRA** to deliver high-quality solutions

EDUCATION

PhD, Computer Science (ongoing)

University of Waterloo

Oct 2022 – Present

- Specialization: Computer Graphics
- Supervisors: Craig S. Kaplan, Lesley Istead

MMath, Computer Science

University of Waterloo

Sep 2021 – Sep 2022

- Specialization: Computer Graphics
- Supervisors: Craig S. Kaplan, Lesley Istead
- Transferred to Ph.D. without completing master's

BCS, Computer Science

University of Waterloo

Sep 2016 – Aug 2021

- Honours with Distinction (cumulative GPA: 84%)
- Co-operative Program
- English Language and Literature Minor

International Baccalaureate Diploma

Cameron Heights Collegiate Institute

Sep 2012 – Sep 2016

- Cumulative GPA: 98%

PUBLICATIONS

Books

- Istead, Lesley, Andreea Pocol, and Sherman Siu (2022b). ***Evaluating Gender Bias in Film Dialogue***. Valencia, Spain: Springer-Verlag, pp. 403–410. ISBN: 978-3-031-08472-0. DOI: 10.1007/978-3-031-08473-7_37.

Journal Articles

- Istead, Lesley, Joe Istead, Andreea Pocol, and Craig S. Kaplan (2022a). "**A Simple, Stroke-Based Method for Gesture Drawing**". In: *Virtual Reality and Intelligent Hardware (VRIH) Journal* 4 (5). Best Paper Award: https://cgs-network.org/legacy/cgi22/CGI22-AWARDS.html. DOI: 10.1016/j.vrih.2022.08.004.

Conference Proceedings

- Pocol, Andreea, Lesley Istead, and Craig Kaplan (2025). "**Binned-KMeans: Extracting Memorable, Explainable, Weighted and Sorted Colour Palettes from Videos**". In: *2025 International Conference on Content-Based Multimedia Indexing (CBMI)*, pp. 1–5. DOI: 10.1109/CBMI66578.2025.11339288.
- Pocol, Andreea, Lesley Istead, and Craig S. Kaplan (2024). "**Evaluating and Improving Disparity Maps Without Ground Truth**". In: *Intelligent Systems and Applications*. Ed. by Kohei Arai. Cham: Springer Nature Switzerland, pp. 153–177. ISBN: 978-3-031-66329-1.
- Pocol, Andreea, Lesley Istead, and Craig Steven Kaplan (2024). "**Evaluating And Improving Disparity Maps Without Ground Truth**". In: *ACM SIGGRAPH 2024 Posters*. SIGGRAPH '24. Denver, CO, USA: Association for Computing Machinery. ISBN: 9798400705168. DOI: 10.1145/3641234.3671049. URL: https://doi.org/10.1145/3641234.3671049.
- Pocol, Andreea, Lesley Istead, Sherman Siu, Sabrina Mokhtari, and Sara Kodeiri (2024). "**Seeing is No Longer Believing: A Survey on the State of Deepfakes, AI-Generated Humans, and Other Non-veridical Media**". In: *Advances in Computer Graphics*. Ed. by Bin Sheng, Lei Bi, Jinman Kim, Nadia Magnenat-Thalmann, and Daniel Thalmann. Cham: Springer Nature Switzerland, pp. 427–440. ISBN: 978-3-031-50072-5.
- Song, Yusen, Andreea Pocol, and Lesley Istead (2024). "**An Evaluation of Gender Bias in 167K Movie Posters**". In: *Intelligent Systems and Applications*. Ed. by Kohei Arai. Cham: Springer Nature Switzerland, pp. 332–358. ISBN: 978-3-031-66329-1.
- Istead, Lesley, Joe Istead, Andreea Pocol, and Craig S. Kaplan (2022b). "**A Simple, Stroke-Based Method for Gesture Drawing**". In: *Computer Graphics International 2022*. Best Paper Award. DOI: 10.1016/j.vrih.2022.08.004.
- Istead, Lesley, Andreea Pocol, and Sherman Siu (2022a). "**Evaluating Gender Bias in Film Dialogue**". In: *Natural Language Processing and Information Systems: 27th International Conference on Applications of Natural Language to Information Systems, NLDB 2022, Valencia, Spain, June 1517, 2022, Proceedings*. Valencia, Spain. DOI: 10.1007/978-3-031-08473-7_37.
- Istead, Lesley, Andreea Pocol, Sherman Siu, William Chen, Alex Zdanowicz, Alex Rowaan, and Craig S. Kaplan (2022). "**The Colour of Horror**". In: *ACM SIGGRAPH European Conference on Visual Media Production (CVMP)*. DOI: 10.1145/3565516.3565523.
- Pocol, Andreea and Lesley Istead (2022). "**Assessing the Impact of Movie Plot Summaries on Box Office Sales**". In: *2022 IEEE Eighth International Conference on Big Data Computing Service and Applications (BigDataService)*, pp. 48–52. DOI: 10.1109/BigDataService55688.2022.00015.
- Istead, Lesley, Andreea Pocol, Craig S. Kaplan, Isaac Watt, Nick Lemoing, and Alicia Yang (2021). "**Generating Rough Stereoscopic 3D Line Drawings from 3D Images**". In: *Proceedings of Graphics Interface 2021*. GI 2021. Virtual Event: Canadian Information Processing Society, pp. 178–185. ISBN: 978-0-9947868-6-9. DOI: 10.20380/GI2021.20.

FEATURED

"Computer science meets cinema: University of Waterloo researchers closely analyze colour in films"

CTV News

March 2023

"Computer Scientists Paint a Picture of Six Decades of Movies"

ACM Tech News

March 2023

Mike Farwell Show (Radio)

Mike Farwell Show

March 2023

- Tuesday March 14th, 2023 [1:25:15]
- Friday March 8th, 2024 [0:18:50]

Four of these faces were produced entirely by AI... can YOU tell who's real? Nearly 40% of people got it wrong in new study

Daily Mail

March 2024

As generative AI becomes more sophisticated, its harder to distinguish the real from the deepfake

The Conversation

March 25, 2025

International Womens Day: Celebrating five Cheriton School of Computer Science researchers

The Conversation

March 8, 2024